



Advanced Time Series Analysis and Forecasting

Seminar 15: Review and project defence



Daniel Traian PELE

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material

- ACI: Adaptive Conformal Inference
- AI: Artificial Intelligence
- AR: AutoRegressive (model); in event studies: Abnormal Return
- BET: Bucharest Exchange Trading (index)
- BNR: Banca Națională a României (the National Bank of Romania)
- CW: Clark–West (test for nested models)
- DM: Diebold–Mariano (test)
- EODHD: EOD Historical Data (market-data provider)
- FRED: Federal Reserve Economic Data
- FWER: Family-Wise Error Rate
- GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- GDP: Gross Domestic Product
- HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- HLN: Harvey–Leybourne–Newbold (small-sample correction of the DM test)
- LR: Likelihood Ratio (test)
- MA: Moving Average
- QML: Quasi-Maximum Likelihood
- RMSE: Root Mean Squared Error
- RMSPE: Root Mean Squared Prediction Error
- SVAR: Structural Vector AutoRegression
- US: United States
- VAR: Vector AutoRegression
- VECM: Vector Error Correction Model

Today's question and route

- **Question:** can you use the tools of the whole course quickly and defend a replication result under questioning?
 - ▶ this seminar comes **before** Lecture 15 and is a project clinic: the section “What you need today” is a formula sheet of the course
- **Route**
 - ▶ Part A: ten short problems from [Chapters 0–14](#), of the kind asked at the defence
 - ▶ Part B: four mock defences on replication outputs (Meese–Rogoff for EUR/RON, the term spread, a leakage audit, a rule search on the BET)
 - ▶ Part C: the pre-registration of your own project; an AI answer to audit
- Notebook for today: [open the seminar notebook in Google Colab](#); each task names its notebook section

Exercise map

Task	Question	Type	Model
A1, A2	the long-run variance of a mean (Ch. 0); a DM test by hand (Ch. 1)	Solved, Proposed	A1
A3, A4	Cholesky responses (Ch. 3); the half-life of an equilibrium error (Ch. 4)	Solved, Proposed	A3
A5, A6	one Kalman step (Ch. 6); a two-state Markov chain (Ch. 7)	Solved, Proposed	A5
A7, A8	the Kupiec test (Ch. 9); GARCH persistence (Ch. 8)	Solved, Proposed	A7
A9, A10	the split-conformal quantile (Ch. 13); placebo p -values and Holm (Ch. 14, 0)	Solved, Proposed	A9
B1, B2	mock defences: Meese–Rogoff for EUR/RON; the term spread	Solved, Proposed	B1
B3, B4	mock defences: a leakage audit; a rule search on the BET	Solved, Proposed	B3
C1, C2	your pre-registration with a power calculation; what is wrong in an AI answer?	Proposed	B1, A1–A10

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model

Data used

Series	Source	Frequency	Period
EUR/RON reference rate	BNR	monthly (end of month)	2005–2026
US real GDP, 10-year and 3-month Treasury yields	FRED	quarterly	1962–2025
BET index	EODHD	daily	2001–2012
Simulated data (leakage audit)	–	–	–

- In the notebook: `dm_test`, `lr_v`, `ols_se`, `leakage_r2`, `stationary_bootstrap_index`, `p_required`; no account or key is needed

What you need today (1/4): inference and forecast evaluation

- Long-run variance: $\Omega = \gamma_0 + 2 \sum_{j \geq 1} \gamma_j$; AR(1): $\gamma_0 = \sigma^2/(1 - \phi^2)$, $\Omega = \sigma^2/(1 - \phi)^2$, $\Omega/\gamma_0 = (1 + \phi)/(1 - \phi)$
 - Newey–West: $\hat{\Omega} = \hat{\gamma}_0 + 2 \sum_{j=1}^L (1 - \frac{j}{L+1}) \hat{\gamma}_j$, $\hat{\gamma}_j = \frac{1}{T} \sum_{t > j} (x_t - \bar{x})(x_{t-j} - \bar{x})$, $L = \lfloor 4(T/100)^{2/9} \rfloor$ Newey and West (1987)
- DM test Diebold and Mariano (1995): $d_t = L(e_{1t}) - L(e_{2t})$, $DM = \bar{d} / \sqrt{\hat{\Omega}_d/P}$; HLN Harvey, Leybourne and Newbold (1997): $DM^* = DM \sqrt{(P+1-2h+h(h-1)/P)/P}$ against t_{P-1}
 - Clark–West for nested models Clark and West (2007): $f_t = e_{1t}^2 - [e_{2t}^2 - (\hat{y}_{1t} - \hat{y}_{2t})^2]$, one-sided test of $\mathbb{E}f_t \leq 0$ ($1 =$ the smaller model)
- Power of a DM test: $P \approx \Omega_d (z_{1-\alpha/2} + z_\pi)^2 / \delta^2$ for a mean gain of δ standard deviations, power π ; AR(1) differential: $\Omega_d = (1 + \rho)/(1 - \rho)$

What you need today (2/4): structure and latent states

- ▣ Recursive SVAR: $u_t = B_0 \varepsilon_t$, $B_0 = \text{chol}(\Sigma_u)$ lower triangular; responses $\Theta_h = A^h B_0$ for a VAR(1)
 - ▶ 2×2 : $b_{11} = \sqrt{\sigma_{11}}$, $b_{21} = \sigma_{21}/b_{11}$, $b_{22} = \sqrt{\sigma_{22} - b_{21}^2}$
- ▣ VECM $\Delta y_t = \alpha \beta' y_{t-1} + \varepsilon_t$: the equilibrium error $z_t = \beta' y_t$ follows $z_t = (1 + \beta' \alpha) z_{t-1} + \beta' \varepsilon_t$; half-life $\ln 0.5 / \ln(1 + \beta' \alpha)$
- ▣ Local level: $y_t = \mu_t + \varepsilon_t$, $\mu_{t+1} = \mu_t + \eta_t$; $F = P + \sigma_\varepsilon^2$, $K = P/F$, $a_{t|t} = a + K(y - a)$, $P_{t|t} = P(1 - K)$, $P_{t+1} = P_{t|t} + \sigma_\eta^2$
 - ▶ steady state: $\bar{P} = \sigma_\varepsilon^2(q + \sqrt{q^2 + 4q})/2$, $q = \sigma_\eta^2/\sigma_\varepsilon^2$
- ▣ Two-state Markov chain: $\mathbb{E}D_i = 1/(1 - p_{ii})$, $\pi_1 = (1 - p_{22})/(2 - p_{11} - p_{22})$, persistence $\lambda = p_{11} + p_{22} - 1$

What you need today (3/4): volatility, risk and conformal prediction

- ▣ GARCH(1,1): $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$; persistence $\alpha + \beta$; half-life $\ln 0.5 / \ln(\alpha + \beta)$; $\sigma^2 = \omega / (1 - \alpha - \beta)$
- ▣ VaR 1%: $\text{VaR}_\alpha = -q_\alpha$; x exceedances in T days; Kupiec: $LR_{uc} = -2 \ln \frac{(1 - \alpha)^{T-x} \alpha^x}{(1 - \hat{\pi})^{T-x} \hat{\pi}^x} \sim \chi_1^2$, $\hat{\pi} = x/T$
 - ▶ Basel traffic light for 250 days at 1%: green 0–4, yellow 5–9, red 10 or more exceedances
- ▣ Split conformal [Romano, Patterson and Candès \(2019\)](#): scores $S_1, \dots, S_n$ on a calibration set; $\hat{q} = S_{(k)}$, $k = \lceil (n+1)(1-\alpha) \rceil$
 - ▶ under exchangeability: $1 - \alpha \leq \Pr(Y_{n+1} \in \hat{C}) \leq 1 - \alpha + 1/(n+1)$ (no ties); ACI [Gibbs and Candès \(2021\)](#): $\alpha_{t+1} = \alpha_t + \gamma(\alpha - \text{err}_t)$

What you need today (4/4): searches, placebos and the defence

- m tests at level α : Bonferroni rejects $p_j \leq \alpha/m$; Holm: order $p_{(1)} \leq \dots \leq p_{(m)}$, reject while $p_{(i)} \leq \alpha/(m - i + 1)$, stop at the first failure
 - ▶ max- t (Reality Check logic [White \(2000\)](#)): compare the best t with the bootstrap distribution of $\max_k t_k^*$, resampled in blocks
- Synthetic control placebo [Abadie, Diamond and Hainmueller \(2015\)](#): $p = \#\{j : r_j \geq r_1\}/(J + 1)$; with J donors the smallest p is $1/(J + 1)$
- Leakage: any choice (selection, scaling, tuning) that uses data after the forecast origin [Kaufman et al. \(2012\)](#)
- At the defence: individual, three blocks of questions (code, results, chapters), no AI tools
 - ▶ a good answer: the number, its uncertainty, the assumption behind it, what would change it

A1: how much does a mean know? (Chapter 0) [Solved]

Task

- ▣ A series is AR(1) with $\phi = 0.6$ and $\sigma^2 = 1$. A team reports $t = 2.5$ for its mean with i.i.d. standard errors.
- ▣ 1. Compute γ_0 and Ω .
- ▣ 2. By what factor is the i.i.d. standard error too small?
- ▣ 3. Compute the corrected t and state the conclusion at 5%.
- ▣ Report: two numbers; one number; one number and one sentence.

Solution

- ▣ 1. $\gamma_0 = 1/(1 - 0.36) = 1.5625$; $\Omega = 1/(1 - 0.6)^2 = 6.25$
- ▣ 2. $\Omega/\gamma_0 = (1 + \phi)/(1 - \phi) = 4$: the standard error is too small by a factor of $\sqrt{4} = 2$
- ▣ 3. $t = 2.5/2 = 1.25 < 1.96$: not significant
- ▣ Interpretation: at the defence, the first question about any t -statistic is "which standard error?"

A2: a Diebold–Mariano test by hand (Chapter 1) [Proposed]

Solution discussed in class

Task

- ▣ Squared-error differentials d_t (benchmark minus model), $h = 1$: 0.8, -0.2, 0.5, 1.1, 0.3, -0.4, 0.9, 0.6, 0.2, 0.4. Model: A1.
- ▣ 1. Compute $\bar{d}$, $\hat{\gamma}_0$ and $\hat{\gamma}_1$ (divide by P).
- ▣ 2. Compute $\hat{\Omega}$ with the Bartlett kernel and $L = 1$, then DM and HLN.
- ▣ 3. Compare with the critical value of t_9 and state the conclusion.
- ▣ Report: three numbers; three numbers; one sentence.

A3: Cholesky responses (Chapter 3) [Solved]

Task

- ▣ A VAR(1) $y_t = Ay_{t-1} + u_t$ with $A = \begin{pmatrix} 0.5 & 0 \\ 0.2 & 0.4 \end{pmatrix}$, $\Sigma_u = \begin{pmatrix} 4 & 2 \\ 2 & 5 \end{pmatrix}$, recursive ordering.
- ▣ 1. Compute B_0 .
 - ▣ 2. Compute $\Theta_1 = AB_0$ and $\Theta_2 = A^2B_0$.
 - ▣ 3. What changes if the ordering is reversed?
 - ▣ Report: a matrix; two matrices; one sentence.

Solution

- ▣ 1. $b_{11} = \sqrt{4} = 2.0$, $b_{21} = 2/2 = 1.0$, $b_{22} = \sqrt{5-1} = 2.0$
- ▣ 2. $\Theta_1 = \begin{pmatrix} 1.00 & 0.00 \\ 0.80 & 0.80 \end{pmatrix}$, $\Theta_2 = \begin{pmatrix} 0.50 & 0.00 \\ 0.52 & 0.32 \end{pmatrix}$
- ▣ 3. Reversed, the first variable responds on impact to the second shock and the second no longer to the first; with $\sigma_{12} \neq 0$ all responses change: the ordering is the identifying assumption
- ▣ Interpretation: a defence question is always “why this ordering?”

A4: the half-life of an equilibrium error (Chapter 4) [Proposed]

Solution discussed in class

Task

- ▣ A bivariate VECM with $\beta = (1, -1)'$ and $\alpha = (-0.2, 0.05)'$, no short-run terms. Model: A3.
- ▣ 1. Derive the AR(1) of $z_t = \beta' y_t$ and its coefficient.
- ▣ 2. Compute the half-life of a deviation.
- ▣ 3. Which variable is weakly exogenous if $\alpha_2 = 0$, and how is that tested?
- ▣ Report: one equation and one number; one number; one sentence.

A5: one step of the Kalman filter (Chapter 6) [Solved]

Task

- ▣ Local level with $\sigma_\varepsilon^2 = 4$, $\sigma_\eta^2 = 1$; predicted state $a = 10$, $P = 2$; observation $y = 13$.
- ▣ 1. Compute v , F and K .
- ▣ 2. Compute $a_{t|t}$, $P_{t|t}$ and P_{t+1} .
- ▣ 3. Compute the steady-state $\bar{P}$ and gain.
- ▣ Report: three numbers; three numbers; two numbers.

Solution

- ▣ 1. $v = 3$, $F = 2 + 4 = 6$, $K = 2/6 = 0.333$
- ▣ 2. $a_{t|t} = 10 + K \cdot 3 = 11$; $P_{t|t} = 2(1 - K) = 1.333$; $P_{t+1} = 2.333$
- ▣ 3. $q = 0.25$: $\bar{P} = 2.562$, $\bar{K} = \bar{P}/(\bar{P} + 4) = 0.390$
- ▣ Interpretation: the filter moves one third of the way towards the surprise; the signal-to-noise ratio q alone sets the long-run gain



A6: a two-state Markov chain (Chapter 7) [Proposed]

Solution discussed in class

Task

- ▣ Regime 1 (expansion) with $p_{11} = 0.95$, regime 2 (recession) with $p_{22} = 0.80$, monthly data. Model: A5.
- ▣ 1. Compute the expected durations.
- ▣ 2. Compute the ergodic probability of regime 1 and the persistence λ .
- ▣ 3. Why can a likelihood-ratio test of one regime against two not use χ^2_2 ?
- ▣ Report: two numbers; two numbers; one sentence.

A7: the Kupiec test for VaR 1% (Chapter 9) [Solved]

Task

- ▣ A VaR 1% model has $x = 6$ exceedances in $T = 250$ days.
- ▣ 1. Compute the hit rate and the expected number of exceedances.
- ▣ 2. Compute LR_{UC} and its p -value; compare with the exact binomial tail $\Pr(X \geq 6)$.
- ▣ 3. Give the Basel zone and the conclusion.
- ▣ Report: two numbers; three numbers; one sentence.

Solution

- ▣ 1. 2.4%; expected 2.5
- ▣ 2. $LR_{UC} = 3.56$, $p = 0.059$; exact one-sided binomial $p = 0.041$
- ▣ 3. Yellow zone; the asymptotic test does not reject at 5%, the exact tail does: with $\alpha = 1\%$ and 250 days, report both
- ▣ Interpretation: backtests on a few tail days have little power; one exceedance changes the verdict

A8: GARCH persistence (Chapter 8) [Proposed]

Solution discussed in class

Task

- ▣ Daily returns in %, GARCH(1,1) with $\omega = 0.02$, $\alpha = 0.08$, $\beta = 0.90$. Model: A7.
- ▣ 1. Compute the persistence and the half-life of a volatility shock.
- ▣ 2. Compute the unconditional variance and the annualised volatility (250 days).
- ▣ 3. Why are Hessian standard errors of α misleading with fat-tailed returns?
- ▣ Report: two numbers; two numbers; one sentence.

A9: the split-conformal quantile (Chapter 13) [Solved]

Task

- ▣ Absolute residuals of 19 calibration points: 0.12, 0.85, 0.40, 1.30, 0.07, 0.66, 2.10, 0.95, 0.31, 1.75, 0.52, 0.22, 1.05, 0.48, 0.90, 1.42, 0.18, 0.73, 1.20; $\alpha = 0.1$.
- ▣ 1. Compute k and $\hat{q}$.
- ▣ 2. Give the coverage bounds of $\hat{y} \pm \hat{q}$.
- ▣ 3. Which assumption fails for daily returns, and what replaces it?
- ▣ Report: two numbers; two numbers; one sentence.

Solution

- ▣ 1. $k = \lceil 20 \cdot 0.9 \rceil = 18$: the 18th smallest score, $\hat{q} = 1.75$
- ▣ 2. Between 0.90 and $0.9 + 1/20 = 0.950$, marginally
- ▣ 3. Exchangeability fails with volatility clustering; ACI or weighted conformal restore long-run coverage, conditional coverage must be tested
- ▣ Interpretation: a guarantee is only as good as its assumption; say which one you rely on

A10: placebo p -values and Holm (Chapters 14 and 0) [Proposed]

Solution discussed in class

Task

- ▣ A synthetic control with 19 donors; the treated unit has the largest post/pre RMSPE ratio. The team tests five outcomes with $p = 0.010, 0.020, 0.040, 0.200, 0.030$. Model: A9.
- ▣ 1. Compute the placebo p -value and the smallest attainable value.
- ▣ 2. How many outcomes are significant at 5% without correction, with Bonferroni and with Holm?
- ▣ 3. What should the pre-registration have said?
- ▣ Report: two numbers; three counts; one sentence.

B1: Meese and Rogoff for EUR/RON [Solved]

- **Question:** a team replicates [Meese and Rogoff \(1983\)](#): can an AR(1) beat the random walk for the monthly EUR/RON change? You are the examiner and the candidate at once
- **Inputs:** BNR reference rate, end of month, $100 \times \log$ change; rolling window of 60 months; 128 one-month forecasts from January 2016
- **Tasks**
 1. Produce the output: RMSE of both models, their ratio, DM–HLN and Clark–West.
 2. Defence question: why is Clark–West the right test here, and what does it test?
 3. Defence question: is the non-rejection evidence that the random walk is best? Use the power formula.
 4. Interpretation: write the one-sentence conclusion you would defend.
- **Report:** four numbers and two test statistics, two short answers, one sentence
- **Notebook:** section B1

B1: solution [Solved]



- RMSE: random walk 0.509, AR(1) 0.524; ratio 1.028; DM–HLN = 1.19 ($p = 0.235$); Clark–West = 1.11 (one-sided $p = 0.134$)
- Clark–West: the random walk is nested in the AR(1); under the null the estimated coefficients add noise to the larger model, which biases DM towards the random walk; CW corrects for it [Clark and West \(2007\)](#)
- Power: first-order autocorrelation of d_t -0.01 ; a gain of 0.2 standard deviations needs about 196 forecasts for 80% power, more than the 128 available
- Interpretation: “the Meese–Rogoff result holds for the leu over 2016–2026: the AR(1) does not beat the random walk, and our sample could only have detected a large gain”

B2: the term spread and growth [Proposed]

- **Question:** a team replicates [Estrella and Hardouvelis \(1991\)](#) and reports “the spread predicts US growth, $t = 4.7$ ”. Defend or correct it
- **Inputs:** FRED real GDP, 10-year and 3-month yields (quarterly averages), 1962–2025; $y_t = (400/4) \ln(\text{GDP}_{t+4}/\text{GDP}_t)$ on the spread; model: B1
- **Tasks**
 1. Estimate the slope with classical, Hansen–Hodrick-type (3 lags) and Newey–West standard errors.
 2. Defence question: why must the errors be at least MA(3)? Report the residual autocorrelation.
 3. Defence question: is the relation stable? Estimate it before 1989 and from 1989.
 4. Interpretation: rewrite the team’s sentence so that it matches the evidence.
- **Report:** a slope, three standard errors and t -statistics, one autocorrelation, two subsample slopes, one sentence
- **Notebook:** section B2

B3: a leakage audit [Solved]

- **Question:** a team selects the 5 best of 100 candidate predictors by their correlation with the target over the whole sample, then reports a positive out-of-sample R^2 on the last 60 periods. Is the gain real?
- **Inputs:** a simulation with pure-noise predictors and target, $T = 240$, 1 000 replications (Lecture 15 design); model: B1
- **Tasks**
 1. Run the team's protocol and the honest protocol (selection on the training sample only).
 2. Report the mean out-of-sample R^2 and the share of positive values for both.
 3. Defence question: which line of code leaks, and how is it fixed?
 4. Interpretation: what would you require before believing a reported R^2 of 10%?
- **Report:** four numbers, one line of code, one sentence
- **Notebook:** section B3

B3: solution [Solved]

- ▣ Team protocol: mean $R^2 = 5.9\%$, positive in 81% of replications (90th percentile 14.4%)
- ▣ Honest protocol: mean $R^2 = -13.3\%$, positive in 8%: the predictors are noise
- ▣ The leak: the correlations are computed on `rows = slice(0, T)`; the fix is `slice(0, n_train)`, and in a rolling design the selection is repeated at every origin
- ▣ Interpretation: a R^2 of 10% is within the range that the leak alone produces; ask for the selection code, a placebo with shuffled targets and a test period never touched

B4: a rule search on the BET [Proposed]

- **Question:** a team reports “the 5-day moving-average rule beats buy-and-hold on the BET, $p = 0.006$ ”. It tried 20 rules. Defend or correct it
- **Inputs:** BET daily log returns, 2001–2012 (2993 days); rules “long if the close is above its n -day average”, $n = 5, 10, \dots, 100$; HAC t of the mean excess return; model: B3
- **Tasks**
 1. Compute the t -statistic of every rule and the naive p -value of the best.
 2. Compute the max- t p -value with the stationary bootstrap (mean block 10 days, 999 resamples).
 3. Defence question: why does the bootstrap resample the 20 series jointly, in blocks?
 4. Interpretation: what would you add before calling it a profitable rule?
- **Report:** 20 statistics, two p -values, one critical value, one sentence
- **Notebook:** section B4

C1: the pre-registration of your project [Proposed]

- **Question:** is your planned evaluation able to answer your question?
- **Inputs:** your team's landmark paper and data; the power formula of today's formula sheet; model: B1
- **Tasks**
 1. Write the seven sections of the pre-registration template (Lecture 15, Appendix) in at most one page.
 2. Write the rule that decides "replicated", "partly" or "not replicated".
 3. Compute the evaluation length needed for 80% power at the gain reported by the paper, with ρ estimated on your training data.
 4. Interpretation: if the needed length exceeds your data, how do you change the question?
- **Report:** one page, one rule, one number, one sentence
- **Notebook:** section C1

C2: audit an AI answer [Proposed]

- ▣ A student asked an AI assistant how to prepare the replication project and its defence. The answer:
- ▣ (a) Use the Diebold-Mariano test to compare your AR(1) with the random walk; it is valid for nested models.
- ▣ (b) With 19 donors, a placebo p-value of 0.01 is achievable if the effect is large.
- ▣ (c) If your result is not significant, the random walk is the true model.
- ▣ (d) For an AR(1) with $\phi = 0.6$, i.i.d. standard errors of the mean are too small by a factor of 2.
- ▣ (e) Standardise the predictors once on the full sample before the walk-forward loop: it saves time and changes nothing.
- ▣ (f) Holm rejects more hypotheses than Bonferroni and controls the same family-wise error rate.
- ▣ (g) You may use the assistant during the oral defence if you declare it in AI_USE.md.
- ▣ Tasks
 - ▶ 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number (notebook section C2).
 - ▶ 2. Report: a list of seven verdicts with one line of justification each.

Key takeaways

- ▣ Every number at the defence comes with its standard error, and the standard error with its assumption
- ▣ Nested models need Clark–West; a non-rejection needs a power calculation
- ▣ Overlapping targets and persistence enlarge standard errors; stability over subsamples is part of the evidence
- ▣ Selection on the whole sample creates out-of-sample gains from noise; a search needs a search-adjusted p -value
- ▣ Write the pre-registration and the verdict rule before the first estimate

After the seminar

- ▣ Lecture 15 develops each topic of today: the course map, one recap per chapter, the research workflow, the replication scoreboard, the failure modes, the oral defence and the AI policy
- ▣ Try the [Proposed] tasks in the notebook
- ▣ C1 is the first draft of your team's proposal and pre-registration
- ▣ Reading: [Clemens \(2017\)](#); [Nosek et al. \(2018\)](#); [Diebold \(2015\)](#)
- ▣ **The seminar is for practice and is not graded; the solutions of [Proposed] tasks are discussed in class**

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